# 1 Vectors in the plane

**Sheet 1** Thu 8 Oct. 2026 *Distance between two points and midpoint of a segment*

1. Warm-up · Quick [CAL] Calculate $(-3)^2$, $(5-(-2))^2$ and $-3^2$.

   **Ans.** 9; 49; $-9$.

   *Prerequisite for calculating a distance: the square of a signed number.*

2. Core · Method [GEO] $A(1,2)$ and $B(4,6)$ in an orthonormal frame. Give the coordinates of $\overrightarrow{AB}$ and then the distance $AB$.

   **Ans.** $\overrightarrow{AB}\begin{pmatrix}3\\4\end{pmatrix}$; $AB=\sqrt{3^2+4^2}=5$.

   *Pitfall: $x_A-x_B$ instead of $x_B-x_A$; forgetting the square root.*

3. Core + · MCQ [REP] The midpoint of $[AB]$ with $A(-2,1)$ and $B(4,-3)$ has coordinates:

   **A.** $(1,-1)$ **B.** $(-1,1)$ **C.** $(2,-2)$ **D.** $(3,-2)$

   **Ans.** A. $(1,-1)$

   *Distractors: half-difference $(3,-2)$; undivided sum $(2,-2)$; sign error $(-1,1)$.*

4. Review · Quick [SUB] Calculate $3x^2-2x+1$ for $x=-2$.

   **Ans.** 17.

5. Thread · Quick [EQ] Solve in $\mathbb{R}$: $3x+5=17$.

   **Ans.** $x=4$

**Sheet 2** Mon 12 Oct. 2026 *Determinant of two vectors; collinearity criterion*

1. Warm-up · Quick [CAL] Calculate $2\times(-3)-4\times5$ and $(-1)\times6-3\times(-2)$.

   **Ans.** $-26$; 0.

   *Determinant skills: cross product with signed numbers.*

2. Core · Method [GEO] Calculate $\det(\vec{u},\vec{v})$ for $\vec{u}\begin{pmatrix}2\\-3\end{pmatrix}$ and $\vec{v}\begin{pmatrix}-4\\6\end{pmatrix}$. What can be concluded?

   **Ans.** $2\times6-(-3)\times(-4)=12-12=0$: $\vec{u}$ and $\vec{v}$ are collinear ($\vec{v}=-2\vec{u}$).

3. Core + · True/false [GEO] True or false? Justify. The vectors $\vec{u}\begin{pmatrix}1\\2\end{pmatrix}$ and $\vec{v}\begin{pmatrix}3\\5\end{pmatrix}$ are collinear.

   **Ans.** False. $\det(\vec{u},\vec{v})=1\times5-2\times3=-1\neq0$.

4. Review · Construction [CON] Draw a circle with centre $O$ and two perpendicular diameters $[AC]$ and $[BD]$. What type of quadrilateral is $ABCD$? Justify.

   **Ans.** A square: diagonals with the same midpoint $O$ (parallelogram), of equal length (rectangle) and perpendicular (rhombus).

5. Thread · Quick [ALG] Programme: choose a number; multiply it by 4; add 5. a. What is the result for 3? b. What is the expression in terms of the starting number $x$?

   **Ans.** a. 17. b. $4x+5$.

**Sheet 3** Tue 13 Oct. 2026 *Collinearity and parallelism: the determinant method*

1. Warm-up · Quick [GEO] Calculate the coordinates of $\overrightarrow{AB}$ for $A(1,-2)$ and $B(4,3)$.

   **Ans.** $\overrightarrow{AB}\begin{pmatrix}3\\5\end{pmatrix}$.

2. Core · Method [GEO] Are the points $A(1,2)$, $B(3,3)$ and $C(7,5)$ collinear?

   **Ans.** $\overrightarrow{AB}\begin{pmatrix}2\\1\end{pmatrix}$, $\overrightarrow{AC}\begin{pmatrix}6\\3\end{pmatrix}$: $\det\left(\overrightarrow{AB},\overrightarrow{AC}\right)=2\times3-1\times6=0$: collinear.

3. Core + · Method [EQ] For what value of $k$ are the vectors $\vec{u}\begin{pmatrix}k\\3\end{pmatrix}$ and $\vec{v}\begin{pmatrix}4\\6\end{pmatrix}$ collinear?

   **Ans.** $\det(\vec{u},\vec{v}) = 6k - 12 = 0$, so $k = 2$.

   *The determinant reduces the problem to a linear equation.*

4. Review · Quick [PRO] Calculate 20% of 350.

   **Ans.** $0.20 \times 350 = 70$.

5. Thread · Translate [LIT] A rectangle has length $x+3$ and width $x$ (in cm). Express its perimeter in terms of $x$ and simplify.

   **Ans.** $P = 2\big((x+3)+x\big) = 4x+6$ (in cm).

**Sheet 4** Thu 15 Oct. 2026 *Problem solving: vector, coordinate or geometric method*

1. Warm-up · Quick [GEO] Complete: $ABCD$ is a parallelogram $\Leftrightarrow \overrightarrow{AB} = \ldots$

   **Ans.** $\overrightarrow{AB} = \overrightarrow{DC}$ (or $\overrightarrow{AD} = \overrightarrow{BC}$).

   *The vertices are named in order: $[AC]$ and $[BD]$ are the diagonals.*

2. Core · Method [REP] $A(0,0)$, $B(4,1)$, $C(5,4)$. Find $D$ so that $ABCD$ is a parallelogram.

   **Ans.** $\overrightarrow{AB} = \overrightarrow{DC}$: $(4,1) = (5-x_D, 4-y_D)$, so $D(1,3)$.

3. Core + · Write-up [DEM] $A(0,0)$, $B(3,1)$, $C(-1,3)$. Prove that $ABC$ is an isosceles right-angled triangle at $A$.

   **Ans.** $AB^2 = 10$, $AC^2 = 10$: $AB = AC$; $BC^2 = 16+4 = 20 = AB^2 + AC^2$: converse of Pythagoras.

   *Two distinct justifications: equal lengths and right angle.*

4. Review · Quick [NUM] Write 450,000 in scientific notation.

   **Ans.** $4.5 \times 10^5$.

5. Thread · Quick [SOL] How many faces, vertices and edges does a cube have?

   **Ans.** 6 faces, 8 vertices, 12 edges.